Unit: Unit 4: Absolute Value **Name:** ______________
Topic: Absolute Value Inequalities and Their Applications **Date:** __________

Advanced (Habanero)

Q1. Solve the inequality $|x| < 3$
(A) $(-3, 3)$
(B) $[-3, 3]$
(C) $(-infty, -3)cup(3, infty)$
(D) $(-infty, 3]$
(E) $[3, infty)$

Q2. Solve the inequality $|2x - 1| > 5$
(A) $(-infty, -2)cup(3, infty)$
(B) $(-2, 3)$
(C) $(-infty, 2]$
(D) $[2, infty)$
(E) $(-2, 2)$

Q3. The absolute value of x is less than 2. What is the solution set?
(A) $(-infty, -2)cup(2, infty)$
(B) $(-2, 2)$
(C) $(-2, 2]$
(D) $[-2, 2]$
(E) $[-2, 2)$

Q4. What is the graph of $|x|geq1$ on a number line?
(A) $(-infty, -1]cup[1, infty)$
(B) $(-1, 1)$
(C) $(-1, 1]$
(D) $[-1, 1]$
(E) $(-infty, 1)$

Q5. Solve $|3x| < 12$
(A) $(-4, 4)$
(B) $(-3, 3)$
(C) $(-2, 2)$
(D) $(-infty, -4]$
(E) $[4, infty)$

Q6. The inequality $|2x + 5|leq3$ is equivalent to
(A) $-4leqxleq-1$
(B) $-1leqxleq4$
(C) $-1leqxleq-4$
(D) $(-infty, -4]cup[4, infty)$
(E) $(-4, -1)$

Q7. Find the solution set of $|x^2 - 4|geq3$
(A) $(-infty, -1]cup[1, infty)$
(B) $(-infty, -1)cup(1, infty)$
(C) $[-1, 1]$
(D) $(-infty, 1]$
(E) $[1, infty)$

Q8. Solve $|x - 2| > 2$
(A) $(-infty, 0)cup(4, infty)$
(B) $(-infty, 0]cup[4, infty)$
(C) $(-infty, 4)$
(D) $[0, 4]$
(E) $(-infty, 4]$

Open Ended Questions (FRQ)

Q9. A company packages coffee in bags that weigh within 0.1 pounds of the advertised weight of 2 pounds. Using absolute value inequalities, find the range of possible weights of the bags and express your answer in interval notation. Give your reasoning.

Q10. The temperature in a greenhouse is supposed to be 75^circF, but it can vary by at most 3^circF. Write an absolute value inequality that represents this situation and graph the solution on a number line. Explain your steps.

Chef's Tips

- Chef's Tips
- Read each question carefully and choose the correct answer.
- For FRQs, make sure to show all your work and justify your reasoning.